

12/09/2025

Clase #19

Mecánica Teórica

12/09/2025

espacio de configuraciones
 2.1 a 2.5

La lagrangiana es una función de las
 coordenadas generalizadas

La lagrangiana no es única

entregar las primeras

$L' = L + \frac{dF(q_i, t)}{dt}$ → demostrarla paso a paso.

La lagrangiana no es única - siempre es $L = T - U$ para una partícula libre.

$L = \frac{1}{2} m \dot{x}^2$

$P_x = \frac{\partial L}{\partial \dot{x}} = m \dot{x}$

$P_i = \frac{\partial L}{\partial \dot{q}_i}$

$\frac{d}{dt} \left(\frac{\partial L}{\partial \dot{q}_i} \right) - \frac{\partial L}{\partial q_i} = 0, \quad i = 0, 1, 2, \dots, n$

$\frac{dP_i}{dt} = \frac{\partial L}{\partial q_i}$

$m \frac{d^2 \vec{r}}{dt^2} = \vec{F} \Rightarrow \frac{d\vec{p}}{dt} = \vec{F} \Rightarrow \frac{dP_i}{dt} = F_i$

$\vec{p} = p_1 \hat{e}_1 + p_2 \hat{e}_2 + p_3 \hat{e}_3 = \sum_{i=1}^3 p_i \hat{e}_i$

$\frac{dP_i}{dt} = \frac{\partial L}{\partial q_i}$

$\frac{\partial P_1}{\partial t} = \frac{\partial L}{\partial q_1}$

$\frac{\partial P_2}{\partial t} = \frac{\partial L}{\partial q_2}$

$\frac{\partial P_n}{\partial t} = \frac{\partial L}{\partial q_n}$

$\frac{\partial L}{\partial q_1} = 0$

$\{q_1, q_2, \dots, q_n\}$

$$\frac{1}{2} m (\dot{r}^2 + r^2 \dot{\theta}^2 + r^2 \sin^2 \theta \dot{\phi}^2) \quad p_i = \frac{\partial L}{\partial \dot{q}_i}$$

$$\frac{d}{dt} \left(\frac{\partial L}{\partial \dot{q}_i} \right) - \frac{\partial L}{\partial q_i} = 0, \quad i=1, 2, \dots, n.$$

$$\boxed{\frac{dp_i}{dt} = \frac{\partial L}{\partial q_i}}$$

$$m \frac{d\vec{v}}{dt} = \vec{F} \Rightarrow \frac{d\vec{p}}{dt} = \vec{F} \Rightarrow \frac{dp_i}{dt} = F_i$$

$$\vec{p} = p_1 \hat{e}_1 + p_2 \hat{e}_2 + p_3 \hat{e}_3 = \sum_{i=1}^3 p_i \hat{e}_i$$

$$\frac{dp_i}{dt} = \frac{\partial L}{\partial q_i}$$

$$\frac{dp_1}{dt} = \frac{\partial L}{\partial q_1}$$

$$\frac{dp_2}{dt} = \frac{\partial L}{\partial q_2}$$

$$\vdots$$

$$\frac{dp_n}{dt} = \frac{\partial L}{\partial q_n}$$

$$\frac{dp_i}{dt} = 0 \Rightarrow p_i = \text{cte}$$

$$\left\{ q_1, q_2, \dots, q_n \right\}$$

$$\boxed{\frac{\partial L}{\partial q_i} = 0}$$

$$L = T = \frac{1}{2} m (\dot{r}^2 + r^2 \dot{\theta}^2 + r^2 \sin^2 \theta \dot{\phi}^2)$$

$$\frac{\partial L}{\partial \phi} = 0; \quad p_\phi \equiv \frac{\partial L}{\partial \dot{\phi}} = m r^2 \sin^2 \theta \dot{\phi}$$

$$\vdots$$

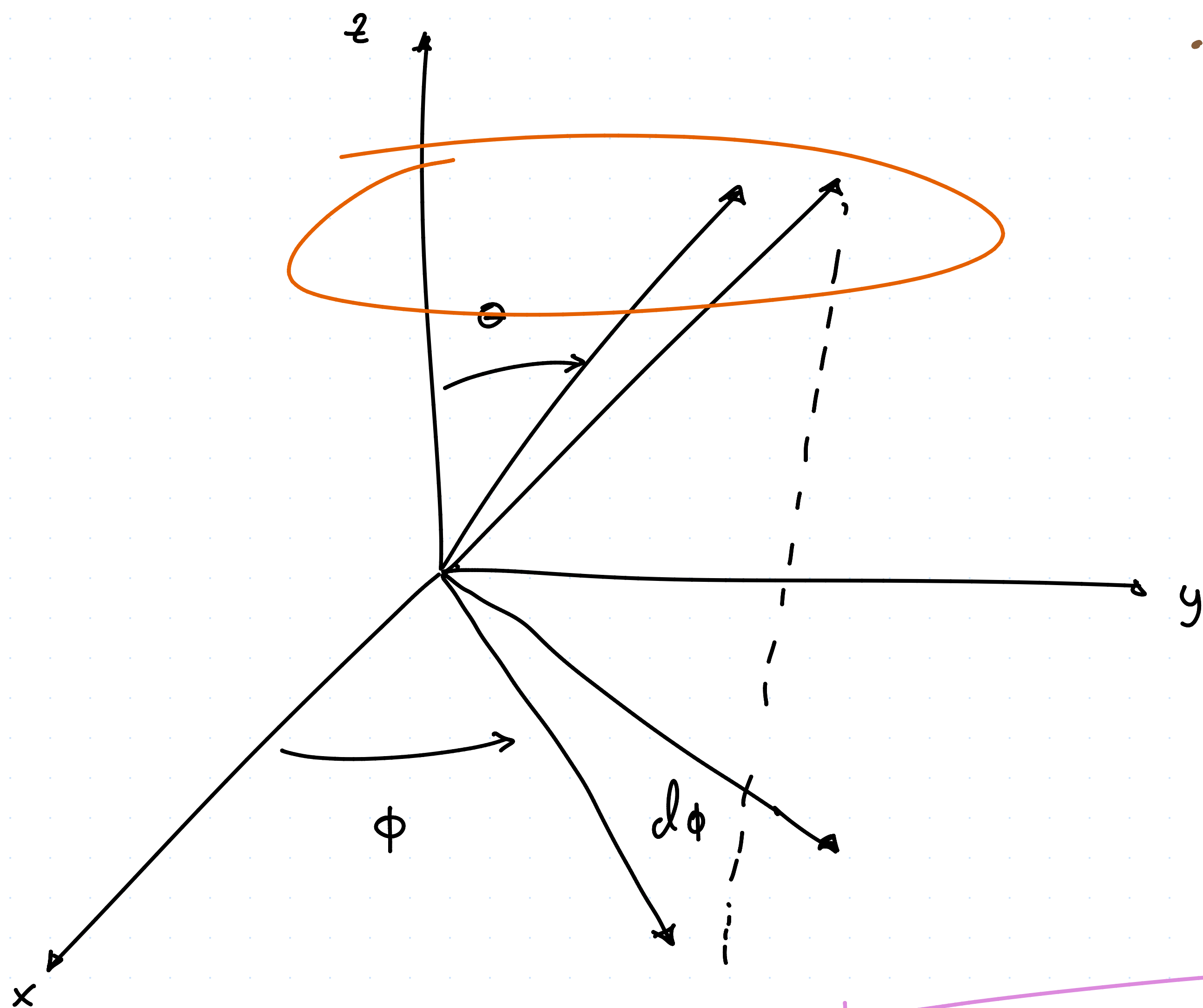
$$= m \dot{x}$$

Teorema

$$\text{s.i.} \quad \frac{\partial L}{\partial q_i} = 0, \quad q_i, \quad \hat{n}$$

$$U = U(q_1, \dots, q_n, \dot{q}_2, \dots, \dot{q}_n)$$

$$p_i = \frac{\partial L}{\partial \dot{q}_i} = \vec{p} \cdot \hat{n} = \text{cte.}$$

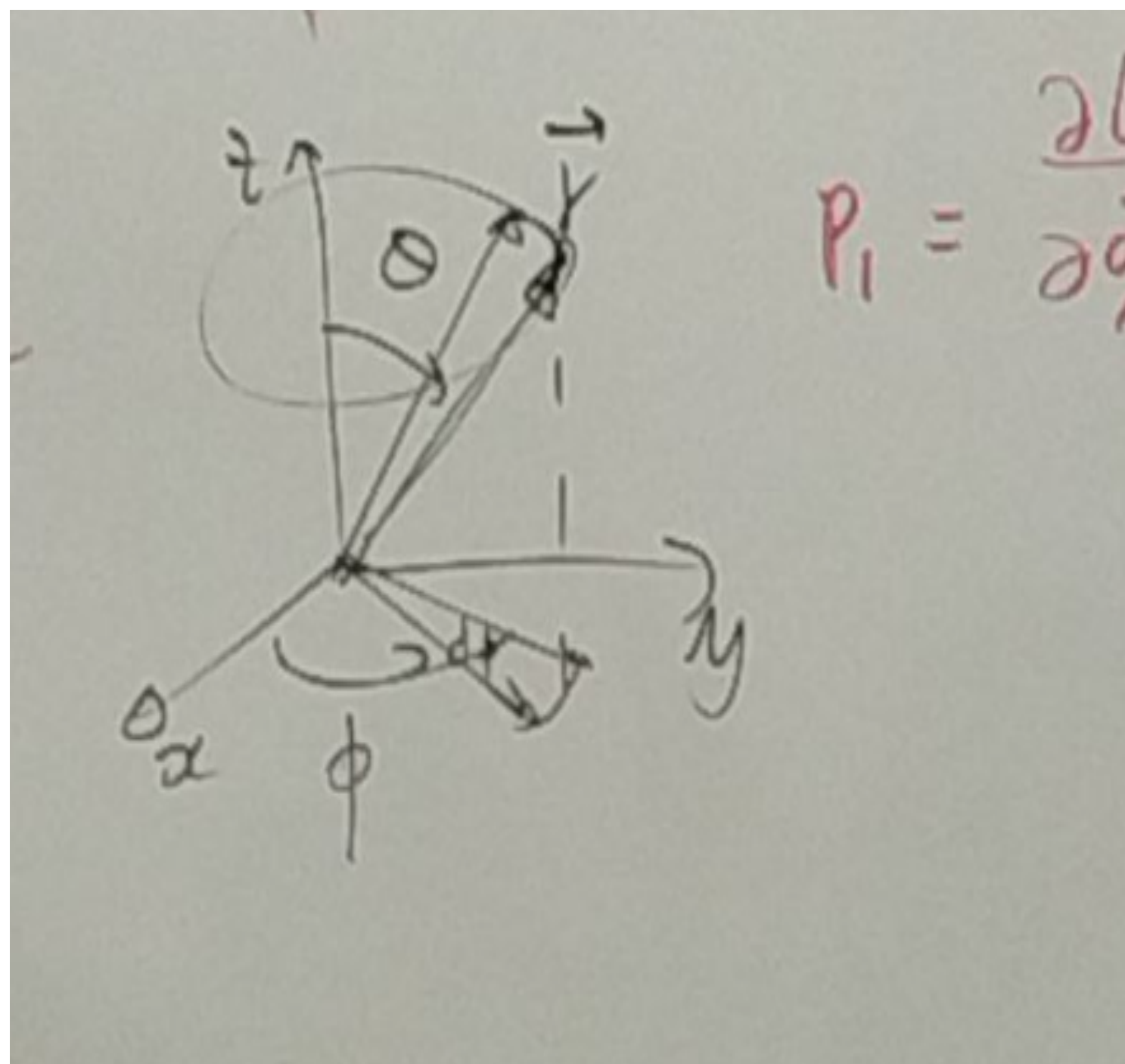


• Momento angular

↳ Invariante ante rotaciones a lo largo del plano

• momento lineal

↳ Invariante ante traslaciones a lo largo del plano



$$\frac{\partial \vec{r}_a}{\partial q_i} = \hat{n}$$

$$\vec{r}_a = \hat{n} q_i + \vec{r}'_a(q_2, \dots, q_n)$$

$$\vec{r} = x\hat{i} + y\hat{j} + z\hat{k}$$

$$\frac{\partial \vec{r}}{\partial x} = \hat{i}$$

$$\dot{p}_i = \frac{\partial L}{\partial \dot{q}_i} ; \quad \frac{\partial U}{\partial \dot{q}_i} = 0$$

Tenemos

$$p_i = \frac{\partial L}{\partial \dot{q}_i} = \frac{\partial}{\partial \dot{q}_i} (T - U)$$

$$= \frac{\partial T}{\partial \dot{q}_i} - \frac{\partial U}{\partial \dot{q}_i} = \frac{\partial T}{\partial \dot{q}_i}$$

Por hipótesis
del problema (vale 0)

$$= \frac{\partial}{\partial \dot{q}_i} \sum_{a=1}^N \frac{1}{2} m_a \dot{\vec{r}}_a^2$$

$$= \sum_{a=1}^N \frac{1}{2} m_a \frac{\partial}{\partial \dot{q}_i} \dot{\vec{r}}_a^2 = \sum_{a=1}^N \cancel{\frac{1}{2} m_a} \cancel{2 \dot{\vec{r}}_a} \cdot \frac{\partial \dot{\vec{r}}_a}{\partial \dot{q}_i}$$

Entonces

$$\boxed{\vec{r}_a = \hat{n} q_i + \vec{r}'_a(q_2, \dots, q_n)}$$

$$\dot{\vec{r}}_a = \hat{n} \dot{q}_i + \dot{\vec{r}}'_a$$

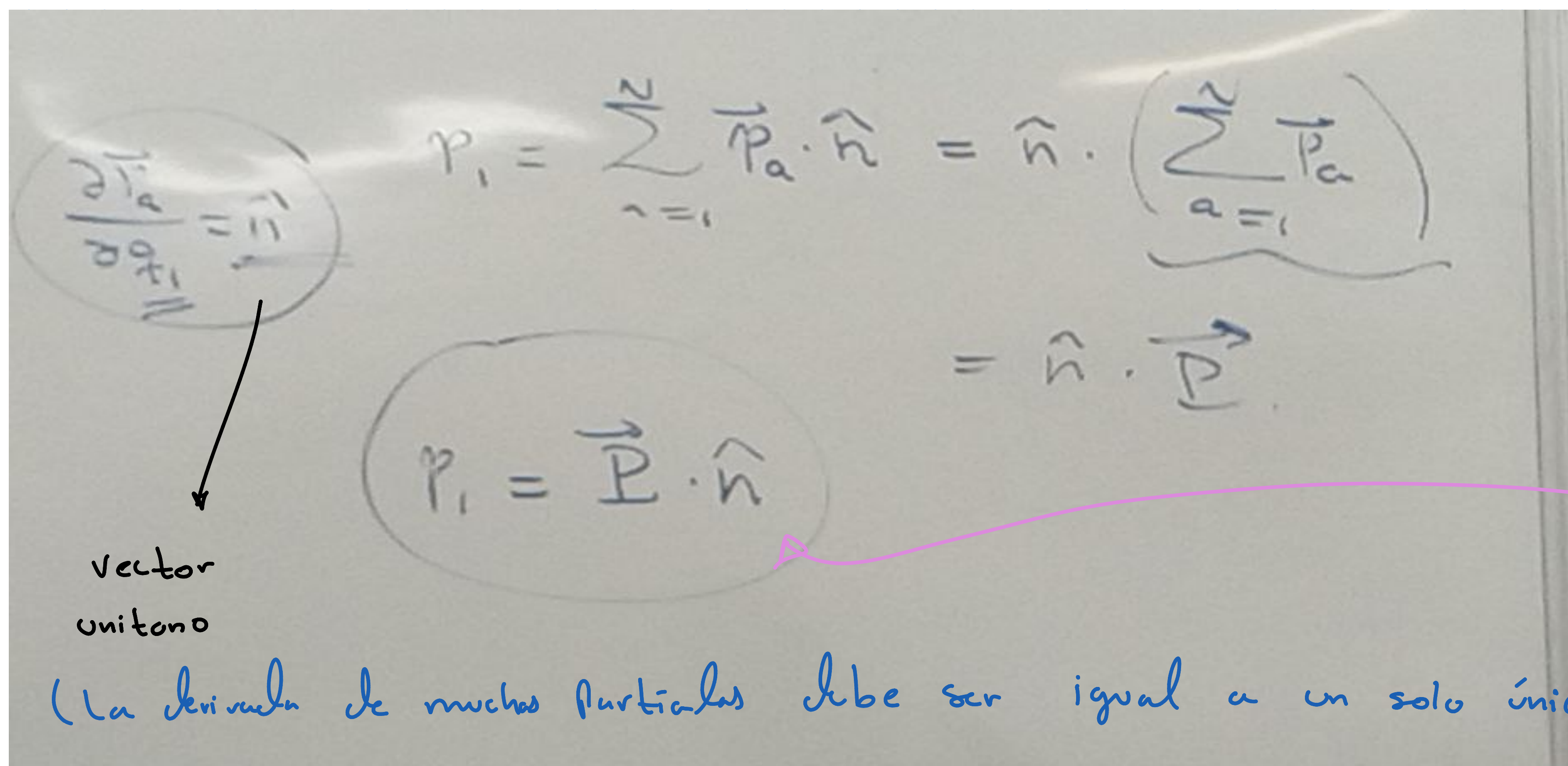
$$\frac{\partial \dot{\vec{r}}_a}{\partial \dot{q}_i} = \hat{n} \frac{\partial \dot{q}_i}{\partial \dot{q}_i} + \cancel{\frac{\partial \dot{\vec{r}}'_a}{\partial \dot{q}_i}} = \hat{n}$$

$$= \sum_{a=1}^N \vec{p}_a \cdot \hat{n}$$

$$p_i = \sum_{a=1}^N \vec{p}_a \cdot \hat{n} = \hat{n} \cdot \left(\sum_{a=1}^N \vec{p}_a \right)$$

$$\vdots$$

$$= \hat{n} \cdot \vec{p}$$



$$\vec{r} = x\hat{i} + y\hat{j} + z\hat{k}$$

$$\frac{\partial \vec{r}}{\partial x} = \hat{i} \Rightarrow p_x = \vec{p} \cdot \hat{i}$$

$$p_y = \vec{p} \cdot \hat{j}$$

$$\frac{\partial \vec{r}}{\partial y} = \hat{j}$$

Proyección del momento lineal del sistema a lo largo de esa dirección

$$p_1 = \vec{p} \cdot \hat{n}$$

Tenemos

$$p_1 = \frac{\partial L}{\partial \dot{q}_1} = \frac{\partial}{\partial \dot{q}_1} (T - U)$$

$$p_1 = \underline{\underline{\vec{p} \cdot \hat{n}}}$$

$$\dot{p}_1 = \frac{\partial L}{\partial q_1}$$

$$\frac{d}{dt} (\vec{p} \cdot \hat{n}) = \frac{\partial L}{\partial q_1}$$

$$\frac{\partial L}{\partial q_1} = \frac{\partial T}{\partial q_1} - \frac{\partial U}{\partial q_1}$$

$$\frac{\partial T}{\partial q_1} = \frac{\partial}{\partial q_1} \sum_{a=1}^N \frac{1}{2} m_a \dot{\vec{r}}_a^2$$

$$= \sum_{a=1}^N \frac{1}{2} m_a \frac{\partial}{\partial q_1} \dot{\vec{r}}_a^2$$

→ Este resultado nos da info el tipo de coordenada. (Revisar notes)

$$\frac{\partial T}{\partial q_1} = \sum_{a=1}^N \frac{1}{2} m_a 2 \dot{\vec{r}}_a \cdot \frac{\partial \dot{\vec{r}}_a}{\partial q_1}$$

$$\vec{r}_a = q_1 \hat{n} + \vec{r}'_a(q_2, \dots, q_n)$$

$$\dot{\vec{r}}_a = \dot{q}_1 \hat{n} + \dot{\vec{r}}'_a$$

$$\frac{\partial \dot{\vec{r}}_a}{\partial q_1} = \frac{\partial \dot{q}_1 \hat{n}}{\partial q_1} + \frac{\partial \dot{\vec{r}}'_a}{\partial q_1} = \vec{0}$$

• No hay dependencia de q_1

Ahora nos hace falta por calcular: $\frac{\partial U}{\partial q_1}$

Anteriormente esto fue 0.

$$\frac{\partial T}{\partial q_1}$$

$$\frac{\partial U}{\partial q_1}$$

$$= Q_1 =$$

$$\sum_{a=1}^N \vec{F}_a \cdot \frac{\partial \vec{r}_a}{\partial q_1}$$

$$\vdots$$

$$= \sum_{a=1}^N \vec{F}_a \cdot \hat{n}$$

Recordando que $\frac{\partial \vec{r}_a}{\partial q_1} = \hat{n}$

Entonces ...

$$\frac{d}{dt} (\vec{p} \cdot \hat{n}) = \frac{\partial L}{\partial q_1} = \vec{F} \cdot \hat{n}$$

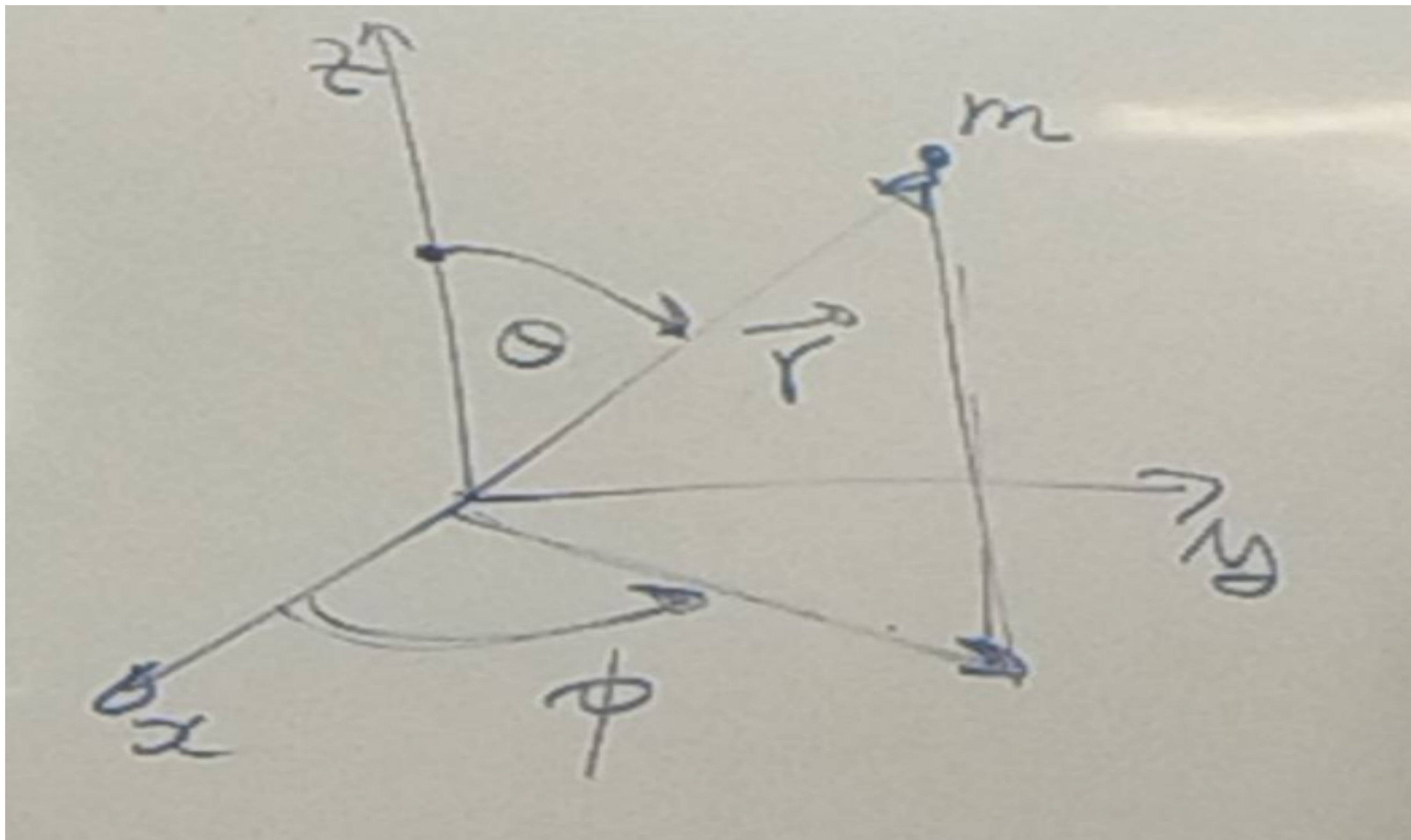
Invarianza del momento angular
orbital

* $\frac{\partial \vec{r}_a}{\partial q_1} = \hat{n}$

* $\frac{\partial \vec{r}_a}{\partial q_2} = ?$

Teorema del momento angular orbital en
donde la coordenada q_2 es cíclica, es
decir no aparece explícitamente en la
Lagrangiana.

Gráfico



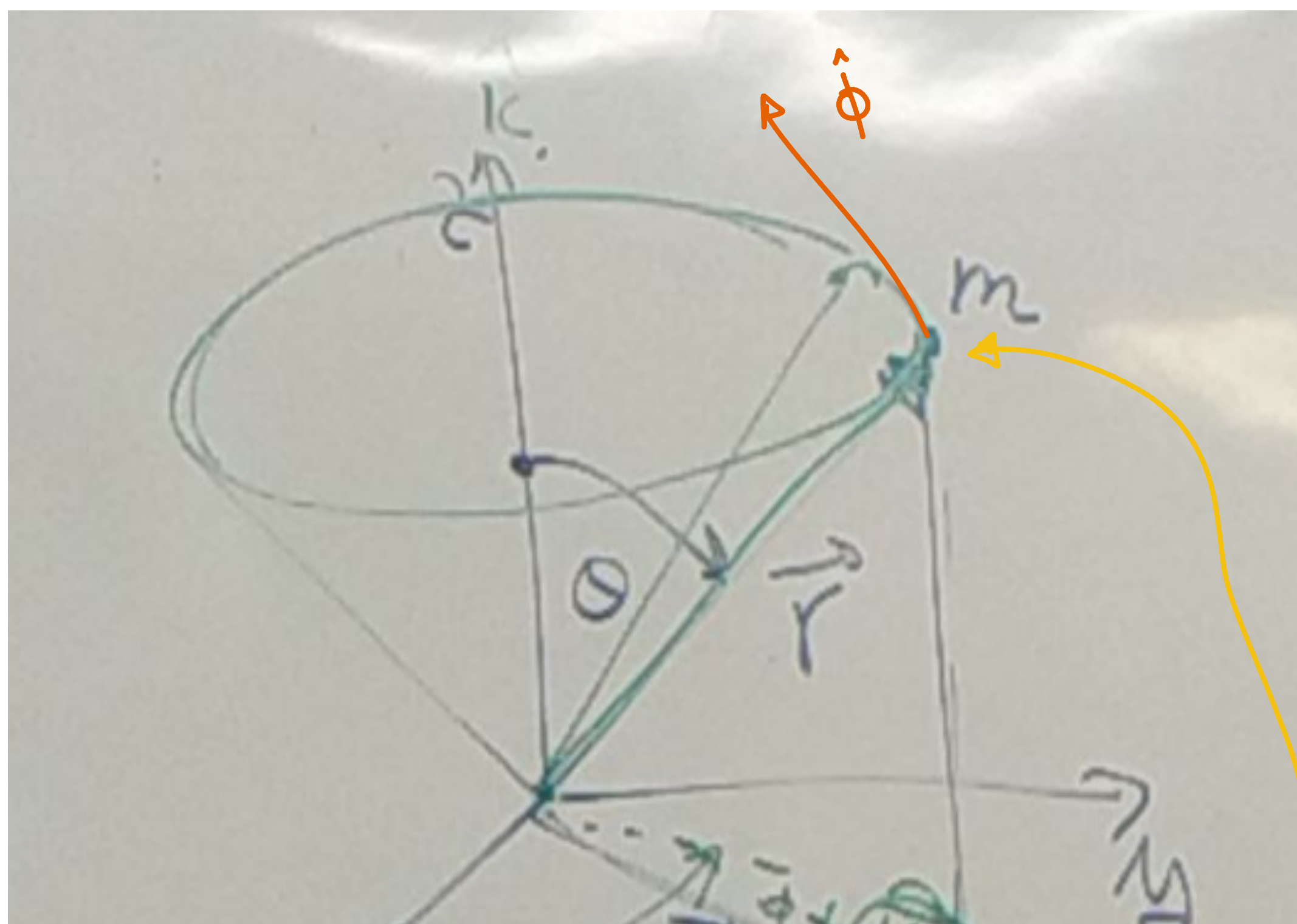
$$\vec{r} = x\hat{i} + y\hat{j} + z\hat{k}$$

$$x = r \cos \phi \sin \theta$$

$$y = r \sin \phi \sin \theta$$

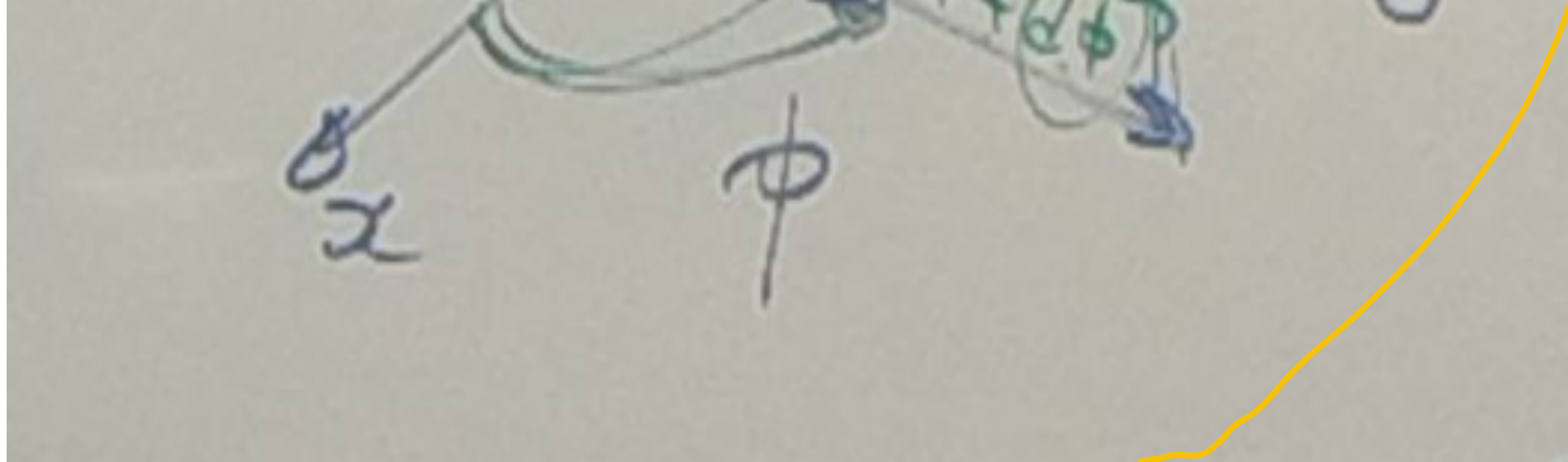
$$z = r \cos \theta$$

$$\vec{r} = r \left[\cos \phi \sin \theta \hat{i} + \sin \phi \sin \theta \hat{j} + \cos \theta \hat{k} \right]$$



$$\frac{\partial \vec{r}}{\partial \phi} = r \left[-\sin \phi \sin \theta \hat{i} + \cos \phi \sin \theta \hat{j} \right]$$

$$\frac{\partial \vec{r}}{\partial \phi} = r \sin \theta \underbrace{\left[-\sin \phi \hat{i} + \cos \phi \hat{j} \right]}_{\hat{\phi}}$$



mantener r constante, theta constante y

cambia phi

phi crece

$$\lim_{\Delta\phi \rightarrow 0} \frac{\vec{r}(\phi + \Delta\phi) - \vec{r}(\phi)}{\Delta\phi}$$

$$\vec{r}(\phi + \Delta\phi) \gg \vec{r}(\phi)$$

La dirección es tangente a la particular.

Tenemos

$$\begin{aligned} \frac{\partial \vec{r}}{\partial \phi} &= r \sin \theta \hat{\phi} \\ &\vdots \\ &= \hat{k} \times \vec{r} \end{aligned}$$

$$\vec{r} = x\hat{i} + y\hat{j} + z\hat{k}$$

$$\frac{\partial \vec{r}}{\partial z} = \hat{k}$$

$$\hat{k} \times \vec{r} = r \left[\cos \phi \sin \theta \hat{k} \times \hat{i} + \sin \phi \sin \theta \hat{k} \times \hat{j} + \cos \theta \hat{k} \times \hat{k} \right]$$

$$\hat{k} \times \vec{r} = r \left[\cos \phi \sin \theta \hat{j} \right]$$

$$- \sin\phi \sin\alpha \hat{z}] \\ = r \sin\alpha [-\sin\phi \hat{z} + \cos\phi \hat{y}]$$

$$\frac{\partial \vec{r}}{\partial \phi} = \hat{k} \times \vec{r} \rightarrow \hat{\phi}$$

$$\frac{\partial \vec{r}}{\partial q_2} = \hat{n} \times \vec{r}$$

El cambio de todos los vectores de posición respecto a la coordenada q_2 de la rotación en la dirección $\hat{n}$ siempre que q_2 es cíclica.

La coordenada q_2 tiene que ver con un ángulo ϕ , tiene que ver con una coordenada cartesiana.

↳ centro de masas (coordenadas adaptadas al centro de masas).

$$\begin{aligned} \vec{r} \times \vec{r} &= r [\cos\phi \sin\alpha \hat{e}_x \hat{z} + \sin\phi \sin\alpha \hat{e}_x \hat{y} + \cos\alpha \hat{e}_x \hat{z}] \\ \vec{r} \times \vec{r} &= r [\cos\phi \sin\alpha \hat{z} - \sin\phi \sin\alpha \hat{y}] \\ &= r \sin\alpha [-\sin\phi \hat{z} + \cos\phi \hat{y}] \\ \frac{\partial \vec{r}}{\partial \phi} &= \hat{k} \times \vec{r} \rightarrow \hat{\phi} \\ \frac{\partial \vec{r}}{\partial q_2} &= \hat{n} \times \vec{r} \end{aligned}$$

$$\frac{\partial \vec{r}}{\partial \phi} = r \sin\alpha [-\sin\phi \hat{z} + \cos\phi \hat{y}]$$

$$\frac{\partial \vec{r}}{\partial \phi} = r \sin\alpha \hat{\phi}$$

$$\frac{\partial \vec{r}}{\partial \phi} = \hat{k} \times \vec{r}$$

$$\frac{\partial \vec{r}}{\partial q_1} = \hat{n}$$

$$\frac{\partial \vec{r}}{\partial q_2} = ?$$

$$\vec{r} = x \hat{e}_1 + y \hat{e}_2 + z \hat{e}_3$$

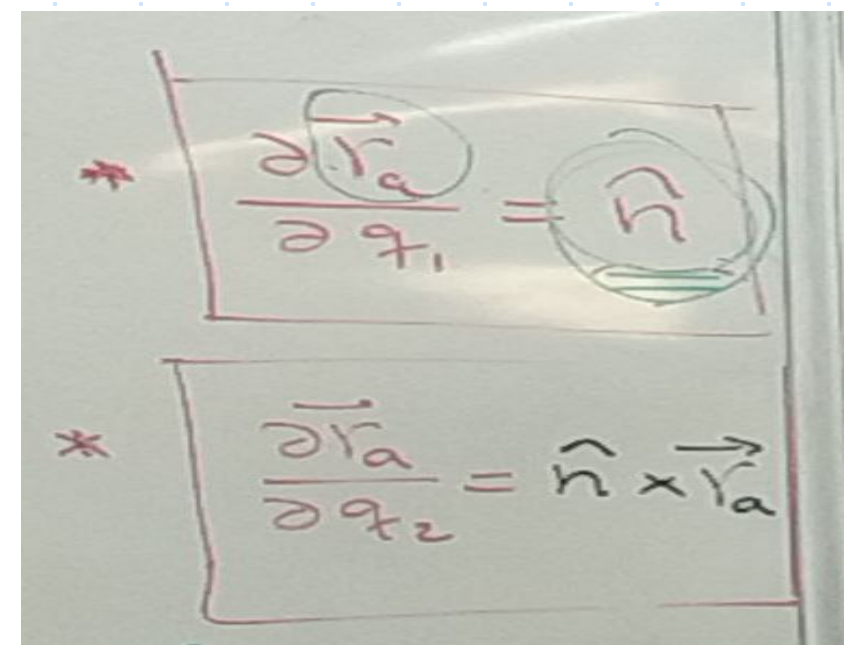
$$\frac{\partial \vec{r}}{\partial \phi} = \hat{k}$$

- Repasar o investigar en estos links

Mostrar:

1:

Está en las notas;
ejercicios



* $\frac{\partial \vec{r}_a}{\partial q_1} = \hat{n}$

* $\frac{\partial \vec{r}_a}{\partial q_2} = \hat{n} \times \vec{r}_a$

$$\left[\vec{r}(r, \theta, \phi + d\phi) \right] \cdot \left[\vec{r}(r, \theta, \phi + d\phi) \right] = \left[\vec{r}(r, \theta, \phi) \right] \cdot \left[\vec{r}(r, \theta, \phi) \right]$$

2:

$$\vec{r} = \vec{r}(r, \theta, \phi)$$

$$\dot{\vec{r}}^2 = \dot{r}^2 + r^2 \dot{\theta}^2 + r^2 \sin^2 \theta \dot{\phi}^2$$

$$\frac{\partial \dot{\vec{r}}^2}{\partial \phi} = 0$$